

JISHNU ROY

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PROFESSIONAL SUMMARY

OCI-certified Backend & AI Systems Engineer specializing in scalable infrastructure, automated data pipelines, and production-level GenAI orchestration. Proficient across Python, Java, and SQL, with a proven track record of deploying low-latency microservices, automating CI/CD workflows, and hosting fine-tuned models on resource-constrained environments.

TECHNICAL SKILLS

Languages: Python, Java, SQL (PostgreSQL, MySQL), JavaScript, Shell Scripting, MATLAB

Frameworks & AI: FastAPI, LangChain, PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, Node.js, React.js

Infrastructure & DevOps: AWS (EC2, S3, RDS, IAM), OCI Architecture, Docker, CI/CD Automation (GitLab CI, Jenkins), Git

Databases & Big Data: MongoDB Atlas, Aurora RDS, Snowflake/SnowSQL (Familiarity)

PROFESSIONAL EXPERIENCE

Radiant Research and Academics Pvt. Ltd.

Kolkata, WB

Backend Engineering Intern / Junior Software Developer

Jan 2026 – Present

- Architected robust backend services on AWS EC2, managing complex data storage and querying with Aurora RDS (PostgreSQL) to support high-volume clinical workflows.
- Optimized production queries and designed normalized ERD frameworks with Role-Based Access Control (RBAC), reducing latency by 40%.
- Engineered automated CI/CD pipelines and configured Nginx reverse proxies, enabling zero-downtime continuous deployment and database migrations.

GenAI Centre of Excellence, IEM

Kolkata, WB

AI Systems Engineering Intern

Dec 2025 – Mar 2026

- Engineered backend architecture for a Natural Language to SQL (NL2SQL) system connecting NLP outputs to structured database queries.
- Orchestrated API integrations and data routing using LangChain and Gemini 2.5 Flash Cloud APIs for high-speed, reliable query generation.
- Deployed and served local SLMs (Qwen2.5-Coder) via Ollama, optimizing inference speed and reducing computational overhead.

IEDC, IEM

Kolkata, WB

Undergraduate Research Trainee (Systems / Embedded)

June 2025 – Nov 2025

- Developed a Virtual Barricading System integrating Facial Recognition and Object Detection utilizing YOLOv5 and OpenCV.
- Deployed optimized PyTorch models on resource-constrained edge devices (Raspberry Pi 4B) to guarantee stable real-time inference.

PROJECTS

EduAi | *Next.js 15, Node.js, TypeScript, Upstash Redis, BullMQ, MongoDB Atlas*

May 2026

- **Architected Multi-Agent AI Loop:** Engineered an asynchronous, self-reflecting LLM state machine using Gemini 2.5 Flash where a Draft Agent generates content, a Critic Agent evaluates it against Bloom's Taxonomy, and a Revision Agent programmatically intercepts and refines failures based on structured JSON schema validation.
- **Engineered High-Availability Redis Bypass:** Designed a zero-downtime, circuit-breaking fallback pipeline for the BullMQ serverless task queue; if the Redis cluster drops offline, the system automatically shifts to inline asynchronous execution, ensuring 100% core feature availability.
- **Optimized Real-Time Event Streaming:** Eliminated static loading bottlenecks by implementing a full-duplex Web-Socket (*wss://*) architecture, broadcasting multi-agent terminal logs, internal pipeline states, and system execution ticks live to the Next.js client application.[cite: 9, 10]

AgriGuru Backend Platform | *Python, FastAPI, MongoDB, AWS, PyTorch*

July 2025 – Present

- Engineered a scalable full-stack backend serving RESTful APIs via FastAPI to manage heavy automated real-time data ingestion pipelines and cloud storage synchronization layers.[cite: 2, 7]
- Integrated an internal computer vision CNN (92% accuracy) and Random Forest models (98% accuracy) into high-availability production cloud environments backed by MongoDB Atlas instances.[cite: 1, 3]

EDUCATION

Institute of Engineering and Management B.Tech in Electronics and Communication Engineering (SGPA / CGPA: 8.68)	Kolkata, WB 2023 – 2027
Burdwan Municipal High School Higher Secondary (Class XII) – WBCHSE (93%)	Burdwan, WB 2019 – 2021
St. Xavier's School Secondary (Class X) – ICSE (94%)	Burdwan, WB 2019

CERTIFICATIONS & AWARDS

- **Winner**, FrostHacks 2026 (1st Prize out of multi-tier tracks under Sustainability and Green Earth theme) Mar 2026
- **Ranked 6th**, Infosys Global Hackathon (Selected out of top 30 global competitive developer cohorts) July 2025
- **Oracle Certified Infrastructure Professional (2025):** Data Science Professional (1Z0-1110-25) & Generative AI Professional (1Z0-1127-25) Nov 2025
- **AWS Cloud Practitioner Essentials** (Coursera Validation) Aug 2025